



[Course](#) > [1. Digital Images](#) > [1.5. Conclusion](#) > 1.5.3. Weekly Graded Assessment

1.5.3. Weekly Graded Assessment

Digital Images

3.0/3.0 points (graded)

Which of the following statements is true?

- ☒ 256 grey values can be addressed with an 8-bit image
- ☐ JPEG compression alters the underlying image data reversibly.
- ☐ Display settings alter the underlying image data.
- ☒ Histograms are a versatile tool to detect saturated images.



Submit

You have used 2 of 2 attempts

Image Metadata Information

2/2 points (graded)

With Fiji, open the "Cell_Colony.jpg" sample and display its metadata window.

What is the "width" of this image in pixel?

406



406

What is the bit depth of the image?



You have used 2 of 2 attempts

✓ Correct (2/2 points)

Image Histogram

2/2 points (graded)

With Fiji, open the "Cell_Colony.jpg" sample and display its histogram.

What is the mode of the pixel intensity distribution?



How many pixels with the mode value are in the image?



You have used 1 of 2 attempts

✓ Correct (2/2 points)

Bit Depth

3.0/3.0 points (graded)

An 8-bit image with a width of 512 pixels and a height of 512 pixels has a size of 256 kByte.

What is the size of the image (in kByte) if the bit depth is changed to 16-bit?

512

✓

512

What is the size of the image (in kByte) if the bit depth is changed to 32-bit?

1024

✓

1024

Submit

You have used 1 of 2 attempts

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